

コンパイラ (3)

2026/10/9

上昇型構文解析

$A \rightarrow \alpha$ の右辺へのマッチングによって A に還元する

出力：最右導出の逆順

入力 (左から) に従って下側から還元

$$\begin{aligned} w &= w_1 w_2 \Leftarrow N_1 w_2 & N_1 \rightarrow w_1 \text{ によって還元} \\ N_1 w_2 &= N_1 w_{21} w_{22} \Leftarrow N_1 N_2 w_{22} \end{aligned}$$

右側の還元が最後になる＝最右導出の逆順

難しい点： $\alpha(i)$ に Σ と N が混在

還元するときのルールの選択がキーポイント

プッシュダウン変換機械 (Pushdown Transducer)

$$T = (\Gamma, \Sigma, O, \Delta, l_0, F)$$

Γ スタック記号の有限集合

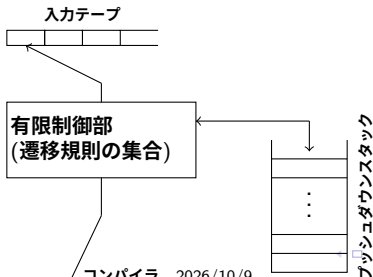
Σ 入力アルファベット

O 出力記号

Δ 遷移関数: $\Gamma^+ \times \Sigma \rightarrow \Gamma^* \times O^* \times \{0, 1\}$

l_0 初期スタック記号

$F \subseteq \Gamma \cup \{\varepsilon\}$ 最終スタック記号集合

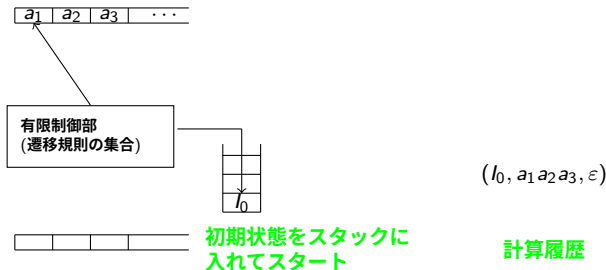


PDT の動作

(スタック, 入力テープ, 出力テープ) : 時点表示 (configuration)

$$\Delta(u, a) = (w, \pi, 1) : u \rightarrow^a (w, \pi)$$

$$\Delta(u, a) = (w, \pi, 0) : u \rightarrow^{(a)} (w, \pi) \quad (\text{先読み動作})$$

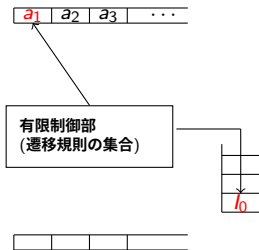


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$$l_0 \rightarrow^{a_1} (c_1 c_2, o_1)$$

$$(l_0, a_1 a_2 a_3, \varepsilon)$$

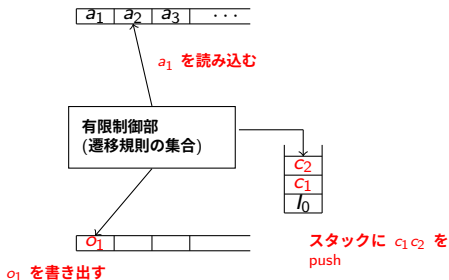
計算履歴

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$$l_0 \rightarrow^{a_1} (c_1 c_2, o_1)$$

$$\vdash (l_0, a_1 a_2 a_3, \varepsilon) \\ (l_0 c_1 c_2, a_2 a_3, o_1)$$

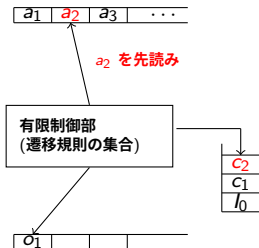
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$$\begin{aligned} l_0 &\rightarrow^{a_1} (c_1 c_2, o_1) \\ c_2 &\rightarrow^{(a_2)} (c_3, o_2) \end{aligned}$$

$$\begin{aligned} &(l_0, a_1 a_2 a_3, \varepsilon) \\ \vdash &(l_0 c_1 c_2, a_2 a_3, o_1) \end{aligned}$$

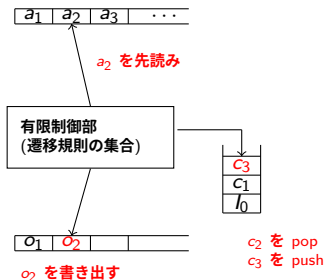
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$$l_0 \rightarrow^{a_1} (c_1 c_2, o_1)$$

$$c_2 \rightarrow^{(a_2)} (c_3, o_2)$$

$$(l_0, a_1 a_2 a_3, \epsilon)$$

$$\vdash (l_0 c_1 c_2, a_2 a_3, o_1)$$

$$\vdash (l_0 c_1 c_3, a_2 a_3, o_1 o_2)$$

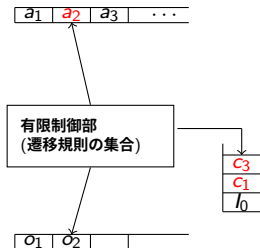
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$$\begin{aligned} l_0 &\rightarrow^{a_1} (c_1 c_2, o_1) \\ c_2 &\rightarrow^{(a_2)} (c_3, o_2) \\ c_1 c_3 &\rightarrow^{a_2} (\varepsilon, o_3) \end{aligned}$$

$$\begin{aligned} &(l_0, a_1 a_2 a_3, \varepsilon) \\ \vdash &(l_0 c_1 c_2, a_2 a_3, o_1) \\ \vdash &(l_0 c_1 c_3, a_2 a_3, o_1 o_2) \end{aligned}$$

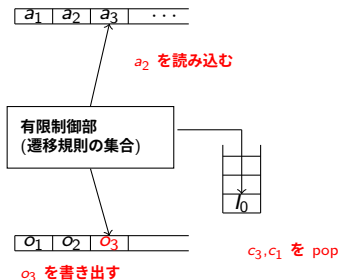
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$$l_0 \rightarrow^{a_1} (c_1 c_2, o_1)$$

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$$(l_0, a_1 a_2 a_3, \varepsilon)$$

$$\vdash (l_0 c_1 c_2, a_2 a_3, o_1)$$

$$\vdash (l_0 c_1 c_3, a_2 a_3, o_1 o_2)$$

$$\vdash (l_0, a_3, o_1 o_2 o_3)$$

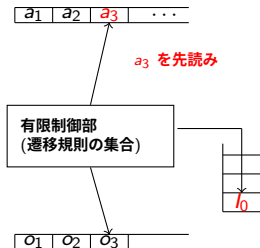
計算履歴

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$$\Delta(u, a) = (w, \pi, 0) : u \rightarrow^{(a)} (w, \pi) \quad (\text{先読み動作})$$



$$\begin{aligned} l_0 &\rightarrow^{a_1} (c_1 c_2, o_1) \\ c_2 &\rightarrow^{(a_2)} (c_3, o_2) \\ c_1 c_3 &\rightarrow^{a_2} (\varepsilon, o_3) \\ l_0 &\rightarrow^{(a_3)} (l_0, \varepsilon) \end{aligned}$$

$$\begin{aligned} &(l_0, a_1 a_2 a_3, \varepsilon) \\ \vdash &(l_0 c_1 c_2, a_2 a_3, o_1) \\ \vdash &(l_0 c_1 c_3, a_2 a_3, o_1 o_2) \\ \vdash &(l_0, a_3, o_1 o_2 o_3) \end{aligned}$$

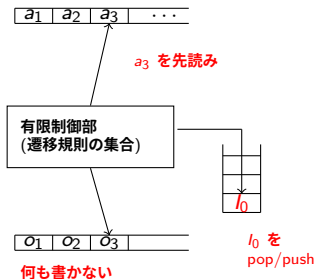
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$$\begin{aligned}
 l_0 &\rightarrow^{a_1} (c_1 c_2, o_1) \\
 c_2 &\rightarrow^{(a_2)} (c_3, o_2) \\
 c_1 c_3 &\rightarrow^{a_2} (\varepsilon, o_3) \\
 l_0 &\rightarrow^{(a_3)} (l_0, \varepsilon) \\
 &(l_0, a_1 a_2 a_3, \varepsilon) \\
 \vdash &(l_0 c_1 c_2, a_2 a_3, o_1) \\
 \vdash &(l_0 c_1 c_3, a_2 a_3, o_1 o_2) \\
 \vdash &(l_0, a_3, o_1 o_2 o_3) \\
 \vdash &(l_0, a_3, o_1 o_2 o_3)
 \end{aligned}$$

計算履歴

構文木の構成と導出

スタック l_0

入力 $(i + n) + i * i \#$

$(i + n) + i * i \#$

構文木の構成と導出

スタック $l_0($

入力 $i + n) + i * i \#$

$(i + n) + i * i \#$

(

$E \rightarrow E + E$ | $E - E$ | $E * E$ | E / E | (E) | i | n | $\#$

構文木の構成と導出

スタック $l_0(i$

入力 $+n) + i * i \#$

$(i + n) + i * i \#$

$(i$

$E \rightarrow E + E$ | $E - E$ | $E * E$ | E / E | (E) | i | n | $\#$

構文木の構成と導出

スタック $l_0(E$

入力 $+n) + i * i\#$

$$(i + n) + i * i\# \Leftarrow_6 (E + n) + i * i\#$$

E
|
(i

$E \rightarrow E + E$ | $E - E$ | $E * E$ | E / E | (E) | i | n | $\#$
 1 2 3 4 5 6 7

構文木の構成と導出

スタック $l_0(E+$

入力 $n) + i * i\#$

$$(i + n) + i * i\# \Leftarrow_6 (E+n) + i * i\#$$

$$\begin{array}{c} E \\ | \\ (\ i \ + \end{array}$$

$E \rightarrow E + E$ | $E - E$ | $E * E$ | E / E | (E) | i | n | $\#$
1 2 3 4 5 6 7

構文木の構成と導出

スタック $l_0(E + n$

入力 $) + i * i \#$

$$(i + n) + i * i \# \leq_6 (E + n) + i * i \#$$

E
|
(i + n

$E \rightarrow E + E$ | $E - E$ | $E * E$ | E / E | (E) | i | n | $\#$
 1 2 3 4 5 6 7

構文木の構成と導出

スタック $l_0(E + E$

入力 $) + i * i \#$

$(i + n) + i * i \# \Leftarrow_6 (E + n) + i * i \#$

$\Leftarrow_7 (E + E) + i * i \#$

$$\begin{array}{cc} E & E \\ | & | \\ (& i + n \end{array}$$

構文木の構成と導出

スタック $l_0(E$

入力 $) + i * i \#$

$$(i + n) + i * i \# \Leftarrow_6 (E + n) + i * i \#$$

$$\Leftarrow_7 (E + E) + i * i \# \Leftarrow_1 (E) + i * i \#$$



$E \rightarrow E + E$ | $E \rightarrow E - E$ | $E \rightarrow E * E$ | $E \rightarrow E / E$ | (E) | i | n

構文木の構成と導出

スタック $l_0(E)$

入力 $+i * i \#$

$$(i + n) + i * i \# \Leftarrow_6 (E + n) + i * i \#$$

$$\Leftarrow_7 (E + E) + i * i \# \Leftarrow_1 (E) + i * i \#$$



$E \rightarrow E + E$ | $E - E$ | $E * E$ | E / E | (E) | i | n

構文木の構成と導出

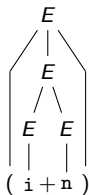
スタック ${}_0E$

入力 $+i * i \#$

$$(i + n) + i * i \# \leftarrow_6 (E + n) + i * i \#$$

$$\leftarrow_7 (E + E) + i * i \# \leftarrow_1 (E) + i * i \#$$

$$\leftarrow_5 E + i * i \#$$



$E \rightarrow E + E$ | $E \rightarrow E - E$ | $E \rightarrow E * E$ | $E \rightarrow E / E$ | (E) | i | n

構文木の構成と導出

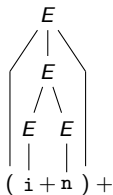
スタック $_0 E +$

入力 $i * i \#$

$$(i + n) + i * i \# \leftarrow_6 (E + n) + i * i \#$$

$$\leftarrow_7 (E + E) + i * i \# \leftarrow_1 (E) + i * i \#$$

$$\leftarrow_5 E + i * i \#$$



$E \rightarrow E + E$ | $E - E$ | $E * E$ | E / E | (E) | i | n

構文木の構成と導出

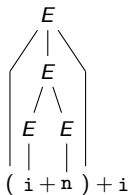
スタック ${}_0E + i$

入力 $*i\#$

$$(i + n) + i * i\# \leftarrow_6 (E+n) + i * i\#$$

$$\leftarrow_7 (E+E) + i * i\# \leftarrow_1 (E) + i * i\#$$

$$\leftarrow_5 E+i * i\#$$



構文木の構成と導出

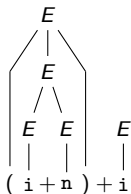
スタック $_0 E + E$

入力 $* i \#$

$$(i + n) + i * i \# \leftarrow_6 (E + n) + i * i \#$$

$$\leftarrow_7 (E + E) + i * i \# \leftarrow_1 (E) + i * i \#$$

$$\leftarrow_5 E + i * i \# \quad \leftarrow_6 E + E * i \#$$



$E \rightarrow E + E$ | $E \rightarrow E - E$ | $E \rightarrow E * E$ | $E \rightarrow E / E$ | (E) | i | n

構文木の構成と導出

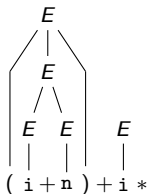
スタック ${}_0E + E*$

入力 $i\#$

$$(i + n) + i * i\# \leftarrow_6 (E+n) + i * i\#$$

$$\leftarrow_7 (E+E) + i * i\# \leftarrow_1 (E) + i * i\#$$

$$\leftarrow_5 E+i * i\# \quad \leftarrow_6 E+E*i\#$$



$E \rightarrow E + E$ | $E \rightarrow E - E$ | $E \rightarrow E * E$ | $E \rightarrow E / E$ | (E) | i | n

構文木の構成と導出

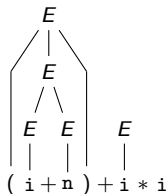
スタック ${}_0 E + E * i$

入力 $\#$

$$(i + n) + i * i \# \leftarrow_6 (E + n) + i * i \#$$

$$\leftarrow_7 (E + E) + i * i \# \leftarrow_1 (E) + i * i \#$$

$$\leftarrow_5 E + i * i \# \quad \leftarrow_6 E + E * i \#$$



$E \rightarrow E + E$ | $E - E$ | $E * E$ | E / E | (E) | i | n

構文木の構成と導出

スタック ${}_0E + E * E$

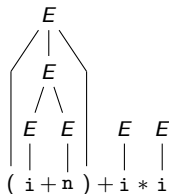
入力 $\#$

$$(i + n) + i * i \# \leftarrow_6 (E + n) + i * i \#$$

$$\leftarrow_7 (E + E) + i * i \# \leftarrow_1 (E) + i * i \#$$

$$\leftarrow_5 E + i * i \# \quad \leftarrow_6 E + E * i \#$$

$$\leftarrow_6 E + E * E \#$$



$E \rightarrow E + E$ | $E \rightarrow E - E$ | $E \rightarrow E * E$ | $E \rightarrow E / E$ | (E) | i | n

構文木の構成と導出

スタック ${}_0E + E$

入力 $\#$

$$(i + n) + i * i\# \leftarrow_6 (E + n) + i * i\#$$

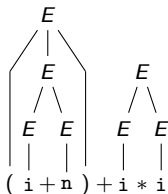
$$\leftarrow_7 (E + E) + i * i\# \leftarrow_1 (E) + i * i\#$$

$$\leftarrow_5 E + i * i\#$$

$$\leftarrow_6 E + E * i\#$$

$$\leftarrow_6 E + E * E\#$$

$$\leftarrow_3 E + E\#$$



$E \rightarrow E + E$ | $E \rightarrow E - E$ | $E \rightarrow E * E$ | $E \rightarrow E / E$ | (E) | i | n | $\#$

構文木の構成と導出

スタック _0E

入力 $\#$

$$(i + n) + i * i \# \leftarrow_6 (E + n) + i * i \#$$

$$\leftarrow_7 (E + E) + i * i \# \leftarrow_1 (E) + i * i \#$$

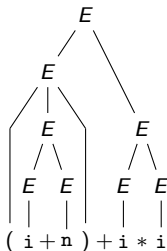
$$\leftarrow_5 E + i * i \#$$

$$\leftarrow_6 E + E * i \#$$

$$\leftarrow_6 E + E * E \#$$

$$\leftarrow_3 E + E \#$$

$$\leftarrow_1 E \#$$



$E \rightarrow E + E$ | $E - E$ | $E * E$ | E / E | (E) | i | n | $\#$

構文木の構成と導出

スタック I_0

入力

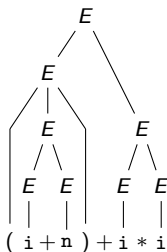
$$(i + n) + i * i \# \leftarrow_6 (E + n) + i * i \#$$

$$\leftarrow_7 (E + E) + i * i \# \leftarrow_1 (E) + i * i \#$$

$$\leftarrow_5 E + i * i \# \quad \leftarrow_6 E + E * i \#$$

$$\leftarrow_6 E + E * E \# \quad \leftarrow_3 E + E \#$$

$$\leftarrow_1 E \#$$



$E \rightarrow E + E$ | $E - E$ | $E * E$ | E / E | (E) | i | n | $\#$

PDT としての動作

$(l_0, (i + n) + i * i\#, \varepsilon)$
 $\vdash (l_0, i + n) + i * i\#, \varepsilon)$
 $\vdash (l_0(i, +n) + i * i\#, \varepsilon)$
 $\vdash (l_0(E, +n) + i * i\#, 6)$
 $\vdash (l_0(E+, n) + i * i\#, 6)$
 $\vdash (l_0(E + n,) + i * i\#, 6)$
 $\vdash (l_0(E + E,) + i * i\#, 67)$
 $\vdash (l_0(E,) + i * i\#, 671)$
 $\vdash (l_0(E), +i * i\#, 671)$
 $\vdash (l_0E, +i * i\#, 6715)$
 $\vdash (l_0E+, i * i\#, 6715)$
 $\vdash (l_0E + i, *i\#, 6715)$
 $\vdash (l_0E + E, *i\#, 67156)$
 $\vdash (l_0E + E*, i\#, 67156)$
 $\vdash (l_0E + E * i, \#, 67156)$
 $\vdash (l_0E + E * E, \#, 671566)$
 $\vdash (l_0E + E, \#, 6715663)$
 $\vdash (l_0E, \#, 67156631)$

(シフトルール)

$\varepsilon \rightarrow ($ $((' , \varepsilon)$
 $\varepsilon \rightarrow)$ $(')', \varepsilon)$
 $\varepsilon \rightarrow i$ $('i', \varepsilon)$
 $\varepsilon \rightarrow +$ $('+', \varepsilon)$
 $\varepsilon \rightarrow *$ $('*', \varepsilon)$

(還元ルール)

$E' + E \rightarrow (E, 1)$
 $E' - E \rightarrow (E, 2)$
 $E' * E \rightarrow (E, 3)$
 $E' / E \rightarrow (E, 4)$
 $'(E)'$ $\rightarrow (E, 5)$
 $i \rightarrow (E, 6)$
 $n \rightarrow (E, 7)$

このままでは、いつどのルールを適用するのがいいのかわからない

上向き構文解析の方針

- ▶ スタックに入力記号を貯める (シフトルール)
- ▶ 生成規則の右辺にマッチしたら、左辺の非終端記号に置き換える。(還元ルール)

上向き構文解析の問題点：還元ルールが適用できるときに還元ルールを適用すべき？

$(i+n)+i+i*i$ の解析において：

$(i+n)+i$ まで解析したとき、規則 1 を適用するか、 $*$ をスタックに push するか？

$w \in L(G)$ という問題では同じだが、構成される構文木が異なる

拡大文法

$S' \notin N$ である非終端記号 S' と生成規則 $\{S' \rightarrow S\}$ を追加
PDT の最終時点表示を明確にする。

先の例

$\vdash (l_0 E, \#, 67156631)$

$\vdash (l_0 S', \#, 67156631)$

遷移規則 $(E \rightarrow^{(\#)} (S', 0))$ を適用

時点表示が $(l_0 S', \#, \dots)$ の形になったら停止

LR 構文解析

PDT を用いた上昇型構文解析：最右導出の逆順を求める

先の例：(無あいまいであれば)13665176 の順で生成規則を適用する最右導出によって解析木を構成可能

$$A \rightarrow \frac{\alpha_1 \cdots \alpha_n}{\text{左からマッチングをとる}}$$

$\alpha_i \in \Sigma$ のとき、入力文字を読みこむ

Shift 動作

$\alpha_i \in N$ のとき、一旦 α をスタックにつんで再帰的にマッチング動作を繰り返す

最終的に α_i がマッチすることがわかれば還元

Reduce 動作

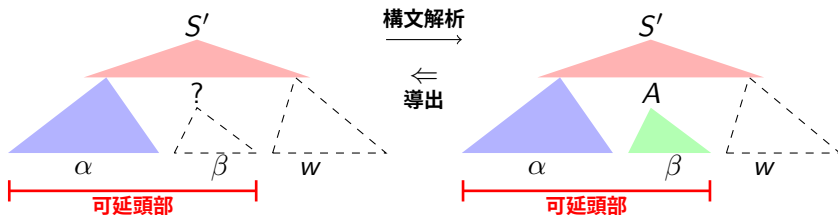
この規則の最後までいけば全体を A に還元する

可延頭部 (viable prefix)

CF 文法 G による最右導出： $S' \Rightarrow_{rm}^* \alpha A w \Rightarrow_{rm} \alpha \beta w$ ($w \in \Sigma^* \#$)

G の可延頭部 (viable prefix) = αA , $\alpha \beta$ の接頭語

一般的な上向き構文解析：可延頭部の木構造を構成していく



右辺が β にマッチする規則を選んで左辺 A に置換

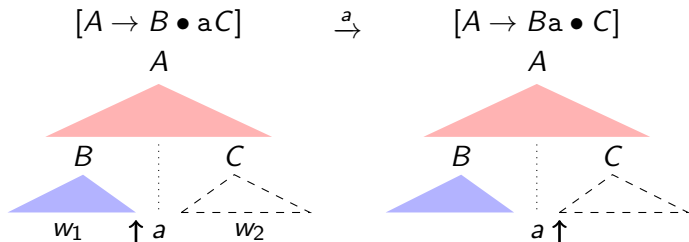
LR(0) 項

各生成規則のどこまでを解析したかマークをつける

$A \rightarrow \alpha_1 \alpha_2 \cdots \alpha_j \cdots \alpha_n$ から以下の LR(0) 項を生成する。

$$\begin{aligned} & [A \rightarrow \bullet \alpha_1 \alpha_2 \cdots \alpha_j \cdots \alpha_n] \\ & [A \rightarrow \alpha_1 \bullet \alpha_2 \cdots \alpha_j \cdots \alpha_n] \\ & \vdots \\ & [A \rightarrow \alpha_1 \alpha_2 \cdots \bullet \alpha_j \cdots \alpha_n] \\ & [A \rightarrow \alpha_1 \alpha_2 \cdots \alpha_j \bullet \cdots \alpha_n] \\ & \vdots \\ & [A \rightarrow \alpha_1 \alpha_2 \cdots \alpha_j \cdots \bullet \alpha_n] \\ & [A \rightarrow \alpha_1 \alpha_2 \cdots \alpha_j \cdots \alpha_n \bullet] \end{aligned}$$

LR(0) 項と構文解析



LR(0) 項間の遷移

LR(0) 項の意味：

その生成規則が適用されて、**•** までの構文解析済みである状態を表す。

構文解析 = LR(0) 項における **• を次に進める**

定義 4.12

LR(0) 項間の遷移関係

$$[A \rightarrow \alpha_1 \alpha_2 \cdots \bullet X \cdots \alpha_n] \xrightarrow{X} [A \rightarrow \alpha_1 \alpha_2 \cdots X \bullet \cdots \alpha_n]$$

$$X \rightarrow \gamma \in P \text{ のとき } [A \rightarrow \beta_1 \bullet X \beta_2] \xrightarrow{\varepsilon} [X \rightarrow \bullet \gamma]$$

LR(0) 項に基づく構文解析の例

$$G_{Arith} = \langle \{S', E, T, F\}, \{i, +, *, (,)\}, P, S' \rangle$$

$$P = \left\{ \begin{array}{l} 0: S' \rightarrow E\# \\ 1: E \rightarrow E + T \\ 2: E \rightarrow T \\ 3: T \rightarrow T * F \\ 4: T \rightarrow F \\ 5: F \rightarrow (E) \\ 6: F \rightarrow i \end{array} \right\}$$

LR(0) 項に基づく構文解析の例

$$G_{Arith} = \langle \{S', E, T, F\}, \{i, +, *, (,)\}, P, S' \rangle$$

LR(0) 項の生成：

$$[S' \rightarrow \bullet E \#], [S' \rightarrow E \bullet \#]$$

$$[E \rightarrow \bullet E + T], [E \rightarrow E \bullet + T], [E \rightarrow E + \bullet T], [E \rightarrow E + T \bullet]$$

$$[E \rightarrow \bullet T], [E \rightarrow T \bullet]$$

$$[T \rightarrow \bullet T * F], [T \rightarrow T \bullet * F], [T \rightarrow T * \bullet F], [T \rightarrow T * F \bullet]$$

$$[T \rightarrow \bullet F], [T \rightarrow F \bullet]$$

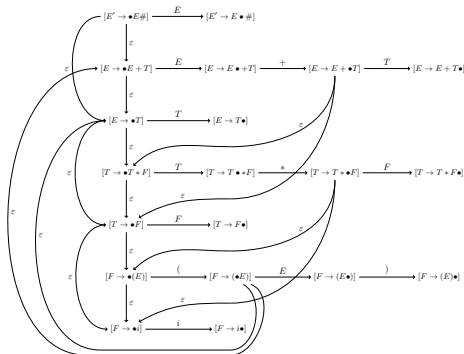
$$[F \rightarrow \bullet (E)], [F \rightarrow (\bullet E)], [F \rightarrow (E \bullet)], [F \rightarrow (E) \bullet]$$

$$[F \rightarrow \bullet i], [F \rightarrow i \bullet]$$

LR(0) 項に基づく構文解析の例

$$G_{Arith} = \langle \{S', E, T, F\}, \{i, +, *, (,)\}, P, S' \rangle$$

遷移規則から LR(0) オートマトンを構成：



NFA なので DFA に変換 可延頭部 α でまとめる

LR(0) 項に基づく構文解析の例

$$G_{Arith} = \langle \{S', E, T, F\}, \{i, +, *, (,)\}, P, S' \rangle$$

正準 LR(0) 項：初期状態から $\alpha = X_1 \cdots X_n$ でたどれる LR 項に一致 ($X_i \in \Sigma \cup N$)

$$\left\{ \begin{array}{l} l_0 = \left\{ [S' \rightarrow \bullet E], [E \rightarrow \bullet E + T], [E \rightarrow \bullet T], [T \rightarrow \bullet T * F], [T \rightarrow \bullet F], \right. \\ \quad \left. [F \rightarrow \bullet (E)], [F \rightarrow \bullet i] \right\}, \\ l_1 = \left\{ [S' \rightarrow E \bullet], [E \rightarrow E \bullet + T] \right\}, \quad l_2 = \left\{ [E \rightarrow T \bullet], [T \rightarrow T \bullet * F] \right\}, \\ l_3 = \left\{ [T \rightarrow F \bullet] \right\}, \\ l_4 = \left\{ [F \rightarrow (\bullet E)], [E \rightarrow \bullet E + T], [E \rightarrow \bullet T], [T \rightarrow \bullet T * F], [T \rightarrow \bullet F], \right. \\ \quad \left. [F \rightarrow \bullet (E)], [F \rightarrow \bullet i] \right\}, \\ l_5 = \left\{ [F \rightarrow i \bullet] \right\}, \\ l_6 = \left\{ [E \rightarrow E + \bullet T], [T \rightarrow \bullet T * F], [T \rightarrow \bullet F], [F \rightarrow \bullet (E)], [F \rightarrow \bullet i] \right\}, \\ l_7 = \left\{ [T \rightarrow T * \bullet F], [F \rightarrow \bullet (E)], [F \rightarrow \bullet i] \right\}, \\ l_8 = \left\{ [F \rightarrow (E \bullet)], [E \rightarrow E \bullet + T] \right\}, \\ l_9 = \left\{ [E \rightarrow E + T \bullet], [T \rightarrow T \bullet * F] \right\}, \\ l_{10} = \left\{ [T \rightarrow T * F \bullet] \right\}, \quad l_{11} = \left\{ [F \rightarrow (E) \bullet] \right\} \end{array} \right\}$$

正準 LR(0) 遷移系の構成

以下の手順に沿って構成される遷移系 $\Theta = (S, \varepsilon)$ を正準 LR(0) 遷移系とよぶ。 S は Θ の状態, ε は Θ の辺である。

ステップ 1: $l_0 = \text{Goto}([S' \rightarrow S\#], \varepsilon)$ とする。

ステップ 2: $\Theta \leftarrow (\{l_0\}, \emptyset); T \leftarrow \{l_0\}$

ステップ 3: $T = \emptyset$ ならば終了する。 Θ に正準 LR(0) 遷移系が得られる。

ステップ 4: $l \in T$ を選び, T から削除する。

ステップ 5: すべての $M \in \text{act}(l)$ に対して,

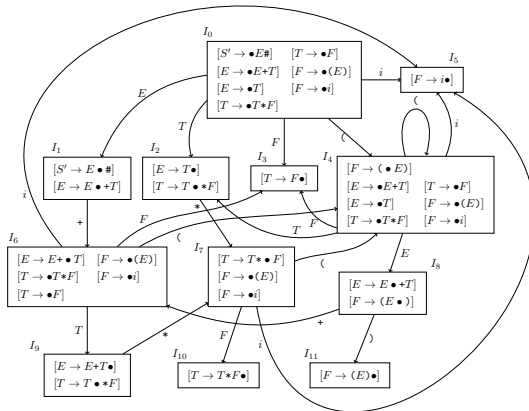
$l' = \{\theta' \mid \theta \in l, \theta \xrightarrow{\varepsilon^*} \xrightarrow{M} \xrightarrow{\varepsilon^*} \theta'\}$ を構成する。

$l' \notin S$ ならば, T に l' を S に l' を追加し, ε に $l \xrightarrow{M} l'$ を追加する。

$l' \in S$ ならば, ε に $l \xrightarrow{M} l'$ を追加する。

ステップ 6: ステップ 3 へもどる。

G_{Arith} の正準 LR(0) 遷移系



正準 LR(0) 遷移系 $\rightarrow$ PDT

$\mathcal{I}$: LR(0) 項全体の集合, $\mathcal{P}(\mathcal{I})$: LR(0) 項集合の集合

$\text{Goto} : \mathcal{P}(\mathcal{I}) \times (N \cup \Sigma)^* \rightarrow \mathcal{I}$

LR(0) オートマトンをたどってたどりつく正準 LR(0) オートマトンの状態

定義 4.13

I を LR(0) 項の集合とする。

$$\text{Goto}(I, \alpha) = \begin{cases} \{\theta' \mid \theta \xrightarrow{\varepsilon}^* \theta' \text{ となる } \theta \in I \text{ が存在する} \} \\ \alpha = \varepsilon \text{ のとき} \\ \text{Goto}(\{\theta' \mid \theta \xrightarrow{M} \theta' \text{ となる } \theta \in \text{Goto}(I, \beta) \text{ が存在する} \}, \alpha) \\ \alpha = \beta M, M \neq \varepsilon \text{ のとき} \end{cases}$$

$\text{Goto}(I, \alpha) = \text{Goto}(I, \alpha\varepsilon) = \text{Goto}(I, \varepsilon\alpha)$ である

可延頭部 α に対して、 $I(\alpha) = \text{Goto}(\{[S' \rightarrow \bullet S\#]\}, \alpha)$

例えば $I(\varepsilon) = I$, $I(E + (E \times E))$ 2026/10/9

PDT の動作

初期スタック: $I(\varepsilon) = I_0$

から始めて、シフトルール、還元ルール、受理ルールを適用

底	$I(\varepsilon)$	X_1	$I(X_1)$	X_2	$I(X_1 X_2)$	$\cdots$	X_n	$I(X_1 \cdots X_n)$	トップ
	状態	記号	状態	記号	状態		記号	状態	

状態と記号を交互にスタックに積む。最後は状態

正準 LR 遷移系 → PDT

スタックトップ $l_{top} = l(X_1 \dots X_n) : LR(0)$ オートマトンの状態

還元 $[A \rightarrow \alpha \bullet] \in l_{top}$ のとき：

$\beta \alpha w \leftarrow_{rm} \beta A w$ (ここで $i : A \rightarrow \alpha$)

- ▶ スタックの上から α にマッチする部分を状態を含めて pop
A を push
- ▶ Goto に従って、 l_ℓ から A で遷移する先の状態を push

$$(l_0 \dots X_{n-k} l_\ell \underbrace{X_{n-k+1} \dots X_n l_n, w\#, o}_{\alpha})$$

$$\vdash (l_0 \dots X_{n-k} l_\ell \underbrace{A}_{\alpha}, w\#, o)$$

$$\vdash (l_0 \dots X_{n-k} l_\ell \underbrace{A l_m}_{\alpha}, w\#, o i)$$

$$l_m = Goto(l_\ell, A)$$

正準 LR 遷移系 $\rightarrow$ PDT

シフト $[A \rightarrow \beta \bullet a\gamma] \in I_{top}$ のとき

$$(I(\varepsilon) \dots X_n I_n, aX, \pi) \vdash (I(\varepsilon) \dots X_n I_n a I_m, X, \pi)$$

ここで、 $I_m = Goto(I_n, a)$

受理 $[S' \rightarrow S \bullet \#] \in I_{top}$ ならば受理

$$(|S'|, \#, \pi) \vdash (acc, \varepsilon, \pi_0)$$

誤り 上記いずれにもあてはまらないとき

SLR(1) 構文解析

ステップ 1: 正準 LR(0) 遷移系 $(S, \mathcal{E})$ を構成し, LR(0) 項 $[S' \rightarrow \bullet S\#]$ を含む S の要素を初期状態 l_0 とする.

ステップ 2': $[A \rightarrow X_1 \cdots X_n \bullet] \in I \in S, a \in \text{Follow}_1(A)$ に対して次の遷移規則を追加する.

$$(IX_1l_1 \cdots X_nl_n, a) \xrightarrow{(a)} (A, r)$$

ここで $l \xrightarrow{X_1} l_1, l_i \xrightarrow{X_i} l_{i+1} \in \mathcal{E}$ であり, r は $A \rightarrow X_1 \cdots X_n$ の生成規則の番号である.

ステップ 3: $l \xrightarrow{M} l' \in \mathcal{E}$ に対して $\theta = [A \rightarrow X_1 \cdots X_i \bullet aX_{i+1} \cdots X_n] \in I, \theta' = [A \rightarrow X_1 \cdots X_i a \bullet X_{i+1} \cdots X_n] \in l'$ のとき次の遷移規則を追加

$$(l_0X_1l_1 \cdots X_il, a) \xrightarrow{a} (l_0X_1l_1 \cdots X_i l a l', \varepsilon)$$

ここで $l_i \xrightarrow{X_{i+1}} l_{i+1} \in \mathcal{E}$ である.

ステップ 4: $[S' \rightarrow S \bullet \#] \in I \in S$ に対して次の遷移規則を追加する.

G_{Arith} による PDT の遷移

$$First_1(E) = First_1(T) = First_1(F) = \{ ' (, i \},$$

$$Follow_1(E) = \{ \#, +, ') \}, Follow_1(T) = Follow_1(F) = \{ \#, +, *, ') \}$$

Shift		Reduce
		$I_0 E I_1 \rightarrow (\#)$ (Acc, ϵ)
		$I_0 E I_1 + I_6 T I_9 \rightarrow (\#, +,)$ ($I_0 E I_1, 1$)
		$I_0 T I_2 \rightarrow (\#, +,)$ ($I_0 E I_1, 2$)
		$I_0 T I_2 * I_7 F I_{10} \rightarrow (\#, *, +,)$ ($I_0 T I_2, 3$)
$I_0 \rightarrow ($	$(I_0(I_4, \epsilon))$	$I_0 F I_3 \rightarrow (\#, *, +,)$ ($I_0 T I_2, 4$)
$I_0 \rightarrow i$	$(I_0 i I_5, \epsilon)$	$I_0(I_4 E I_8) I_{11} \rightarrow (\#, *, +,)$ ($I_0 F I_3, 5$)
$I_1 \rightarrow +$	$(I_1 + I_6, \epsilon)$	$I_0 i I_5 \rightarrow (\#, *, +,)$ ($I_0 F I_3, 6$)
$I_2 \rightarrow *$	$(I_2 * I_7, \epsilon)$	$I_4 E I_8 + I_6 T I_9 \rightarrow (\#, +,)$ ($I_4 E I_8, 1$)
$I_4 \rightarrow ($	$(I_4(I_4, \epsilon))$	$I_4 T I_2 \rightarrow (\#, +,)$ ($I_4 E I_8, 2$)
$I_6 \rightarrow i$	$(I_6 i I_5, \epsilon)$	$I_4 T I_2 * I_7 F I_{10} \rightarrow (\#, *, +,)$ ($I_4 T I_2, 3$)
$I_7 \rightarrow i$	$(I_7 i I_5, \epsilon)$	$I_4 F I_3 \rightarrow (\#, *, +,)$ ($I_4 T I_2, 4$)
$I_7 \rightarrow ($	$(I_7(I_4, \epsilon))$	$I_4(I_4 E I_8) I_{11} \rightarrow (\#, *, +,)$ ($I_4 F I_3, 5$)
$I_8 \rightarrow +$	$(I_8 + I_6, \epsilon)$	$I_4 i I_5 \rightarrow (\#, *, +,)$ ($I_4 F I_3, 6$)
$I_8 \rightarrow)$	$(I_8) I_{11}, \epsilon)$	$I_6 T I_9 * I_7 F I_{10} \rightarrow (\#, *, +,)$ ($I_6 T I_9, 3$)
$I_9 \rightarrow *$	$(I_9 * I_7, \epsilon)$	$I_6 F I_3 \rightarrow (\#, *, +,)$ ($I_6 T I_9, 4$)
		$I_6(I_4 E I_8) I_{11} \rightarrow (\#, *, +,)$ ($I_6 F I_3, 5$)
		$I_6 i I_5 \rightarrow (\#, *, +,)$ ($I_6 F I_3, 6$)
		$I_7(I_4 E I_8) I_{11} \rightarrow (\#, *, +,)$ ($I_7 F I_{10}, 5$)
		$I_7(I_4 E I_8) I_{11} \rightarrow (\#, *, +,)$ ($I_7 F I_{10}, 6$)

SLR(1) 構文解析可能でない場合

- ▶ 先読み文字でどの生成規則で還元するかが特定可能
特定できない場合：Reduce/Reduce conflict
- ▶ 同じ LR 集成に $[A \rightarrow \alpha \bullet a\beta]$ が含まれるとき、 $a \notin Follow_1(A)$
 $a \in Follow_1(A)$ のとき：Shift/Reduce conflict

Reduce 遷移、Shift 遷移が決定的に行われる場合
= **SLR(1) 構文解析可能**

SLR(1) 解析表

PDT の遷移を表で表わす。

Action 表 $(\{I_0, \dots, I_n\} \times \Sigma \cup \{\#\}) + \text{Goto 表 } (\{I_0, \dots, I_n\} \times N)$

Action 表:

(Shift 遷移): $I_i \xrightarrow{a} (I_i a | j, \varepsilon)$ に対して (I_i, a) のエントリに s_j を記入

(Reduce 遷移): $I_{k_0} X_1 I_{k_1} X_2 \cdots X_n I_{k_n} \xrightarrow{(a)} (I_{k_0} A | \ell)$ に対して、
 $A \rightarrow X_1 \cdots X_n$ の規則番号を m とすると (I_{k_n}, a) のエントリに r_m を記入

Goto 表: $I_{k_0} X_1 I_{k_1} X_2 \cdots X_n I_{k_n} \xrightarrow{(a)} (I_{k_0} A | \ell)$ に対して、 (I_{k_0}, A) のエントリに l を記入

G_{Arith} の SLR(1) 解析表

	action						goto		
	i	+	*	(	)	#	E	T	F
0	s5			s4			1	2	3
1		s6				acc			
2		r2	s7		r2	r2			
3		r4	r4		r4	r4			
4	s5			s4			8	2	3
5		r6	r6		r6	r6			
6	s5			s4				9	3
7	s5			s4					10
8		s6			s11				
9		r1	s7		r1	r1			
10		r3	r3		r3	r3			
11		r5	r5		r5	r5			

SLR(1) 解析表にもとづく振舞い

- (1) スタックトップが l_i で、先読み文字 a であるとき、 (l_i, a) に s_j が記入されていれば、スタックに a, l_j を Push

$$l_i \xrightarrow{a} l_i a l_j$$

- (2) スタックトップが l_i で、 $[A \rightarrow X_1 \cdots X_n \bullet] \in l_i$ 、先読み文字 $a \in \text{Follow}_1(A)$ のとき、 $X_1 l_{j_1} X_2 l_{j_2} \cdots X_n l_i$ を pop する。このときスタックトップが l_k となると $A l_\ell$ を push する。(ここで、 j は Goto 表の (l_i, A) に書かれている l_j である。)

G_{Arith} の PDT による解析

$(l_0,$	$i + i * i\#, \varepsilon)$	Action(0, i) = s5
$\vdash (l_0 \ i \ i_5,$	$+i * i\#, \varepsilon)$	Action(5, $+$) = r6, Goto(0, F) = 3
$\vdash (l_0 \ F \ i_3,$	$+i * i\#, 6)$	Action(3, $+$) = r4, Goto(0, T) = 2
$\vdash (l_0 \ T \ i_2,$	$+i * i\#, 64)$	Action(2, $+$) = r2, Goto(0, E) = 1
$\vdash (l_0 \ E \ i_1,$	$+i * i\#, 642)$	Action(1, $+$) = s6
$\vdash (l_0 \ E \ i_1 + i_6,$	$i * i\#, 642)$	Action(6, i) = s5
$\vdash (l_0 \ E \ i_1 + i_6 \ i \ i_5,$	$*i\#, 642)$	Action(5, i) = r6, Goto(6, F) = 3
$\vdash (l_0 \ E \ i_1 + i_6 \ F \ i_3,$	$*i\#, 6426)$	Action(3, $*$) = r4, Goto(6, T) = 9
$\vdash (l_0 \ E \ i_1 + i_6 \ T \ i_9,$	$*i\#, 64264)$	Action(9, $*$) = s7
$\vdash (l_0 \ E \ i_1 + i_6 \ T \ i_9 * i_7 \ i \ i_5,$	$i\#, 64264)$	Action(7, i) = s5
$\vdash (l_0 \ E \ i_1 + i_6 \ T \ i_9 * i_7 \ i \ i_5,$	$\#, 64264)$	Action(5, $\#$) = r6, Goto(7, F) = 10
$\vdash (l_0 \ E \ i_1 + i_6 \ T \ i_9 * i_7 \ F \ i_{10},$	$\#, 642646)$	Action(10, $\#$) = r3, Goto(6, T) = 9
$\vdash (l_0 \ E \ i_1 + i_6 \ T \ i_9,$	$\#, 6426463)$	Action(9, $\#$) = r1, Goto(0, E) = 1
$\vdash (l_0 \ E \ i_1,$	$\#, 64264631)$	Action(1, $\#$) = acc
$\vdash (acc,$	$\#, 64264631)$	

0: $E' \rightarrow E$

1: $E \rightarrow E + T$ 2: $E \rightarrow T$

3: $T \rightarrow T * F$ 4: $T \rightarrow F$

5: $F \rightarrow (E)$ 6: $F \rightarrow i$

G_{Arith} の PDT による解析

$(l_0,$	$i + i * i\#, \varepsilon)$	Action(0, i) = $s5$
$\vdash (l_0 \textcolor{red}{i} \textcolor{red}{l_5},$	$+ i * i\#, \varepsilon)$	Action(5, $+$) = $r6$, Goto(0, F) = 3
$\vdash (l_0 \textcolor{red}{F} \textcolor{red}{l_3},$	$+ i * i\#, 6)$	Action(3, $+$) = $r4$, Goto(0, T) = 2
$\vdash (l_0 \textcolor{red}{T} \textcolor{red}{l_2},$	$+ i * i\#, 64)$	Action(2, $+$) = $r2$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$+ i * i\#, 642)$	Action(1, $+$) = $s6$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6},$	$i * i\#, 642)$	Action(6, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{i} \textcolor{red}{l_5},$	$* i\#, 642)$	Action(5, i) = $r6$, Goto(6, F) = 3
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{F} \textcolor{red}{l_3},$	$* i\#, 6426)$	Action(3, $*$) = $r4$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$* i\#, 64264)$	Action(9, $*$) = $s7$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$i\#, 64264)$	Action(7, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$\#, 64264)$	Action(5, $\#$) = $r6$, Goto(7, F) = 10
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{F} \textcolor{red}{l_{10}},$	$\#, 642646)$	Action(10, $\#$) = $r3$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$\#, 6426463)$	Action(9, $\#$) = $r1$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$\#, 64264631)$	Action(1, $\#$) = acc
$\vdash (acc,$	$\#, 64264631)$	

0: $E' \rightarrow E$

1: $E \rightarrow E + T$ 2: $E \rightarrow T$

3: $T \rightarrow T * F$ 4: $T \rightarrow F$

5: $F \rightarrow (E)$ 6: $F \rightarrow i$

G_{Arith} の PDT による解析

$(l_0,$	$i + i * i\#, \varepsilon)$	Action(0, i) = $s5$
$\vdash (l_0 \textcolor{red}{i} \textcolor{red}{l_5},$	$+i * i\#, \varepsilon)$	Action(5, $+$) = $r6$, Goto(0, F) = 3
$\vdash (l_0 \textcolor{red}{F} \textcolor{red}{l_3},$	$+i * i\#, 6)$	Action(3, $+$) = $r4$, Goto(0, T) = 2
$\vdash (l_0 \textcolor{red}{T} \textcolor{red}{l_2},$	$+i * i\#, 64)$	Action(2, $+$) = $r2$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$+i * i\#, 642)$	Action(1, $+$) = $s6$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6},$	$i * i\#, 642)$	Action(6, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{i} \textcolor{red}{l_5},$	$*i\#, 642)$	Action(5, i) = $r6$, Goto(6, F) = 3
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{F} \textcolor{red}{l_3},$	$*i\#, 6426)$	Action(3, $*$) = $r4$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$*i\#, 64264)$	Action(9, $*$) = $s7$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$i\#, 64264)$	Action(7, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$\#, 64264)$	Action(5, $\#$) = $r6$, Goto(7, F) = 10
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{F} \textcolor{red}{l_{10}},$	$\#, 642646)$	Action(10, $\#$) = $r3$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$\#, 6426463)$	Action(9, $\#$) = $r1$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$\#, 64264631)$	Action(1, $\#$) = acc
$\vdash (acc,$	$\#, 64264631)$	

- | | |
|--------------------------|----------------------|
| 0: $E' \rightarrow E$ | |
| 1: $E \rightarrow E + T$ | 2: $E \rightarrow T$ |
| 3: $T \rightarrow T * F$ | 4: $T \rightarrow F$ |
| 5: $F \rightarrow (E)$ | 6: $F \rightarrow i$ |

G_{Arith} の PDT による解析

$(l_0,$	$i + i * i\#, \varepsilon)$	Action(0, i) = $s5$
$\vdash (l_0 \textcolor{red}{i} \textcolor{red}{l_5},$	$+ i * i\#, \varepsilon)$	Action(5, $+$) = $r6$, Goto(0, F) = 3
$\vdash (l_0 \textcolor{red}{F} \textcolor{red}{l_3},$	$+ i * i\#, 6)$	Action(3, $+$) = $r4$, Goto(0, T) = 2
$\vdash (l_0 \textcolor{red}{T} \textcolor{red}{l_2},$	$+ i * i\#, 64)$	Action(2, $+$) = $r2$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$+ i * i\#, 642)$	Action(1, $+$) = $s6$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6},$	$i * i\#, 642)$	Action(6, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{i} \textcolor{red}{l_5},$	$* i\#, 642)$	Action(5, i) = $r6$, Goto(6, F) = 3
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{F} \textcolor{red}{l_3},$	$* i\#, 6426)$	Action(3, $*$) = $r4$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$* i\#, 64264)$	Action(9, $*$) = $s7$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$i\#, 64264)$	Action(7, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$\#, 64264)$	Action(5, $\#$) = $r6$, Goto(7, F) = 10
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{F} \textcolor{red}{l_{10}},$	$\#, 642646)$	Action(10, $\#$) = $r3$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$\#, 6426463)$	Action(9, $\#$) = $r1$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$\#, 64264631)$	Action(1, $\#$) = acc
$\vdash (acc,$	$\#, 64264631)$	

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1: $E \rightarrow E + T$ 2: $E \rightarrow T$

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G_{Arith} の PDT による解析

$(l_0,$	$i + i * i\#, \varepsilon)$	Action(0, i) = $s5$
$\vdash (l_0 \textcolor{red}{i} \textcolor{red}{l_5},$	$+ i * i\#, \varepsilon)$	Action(5, $+$) = $r6$, Goto(0, F) = 3
$\vdash (l_0 \textcolor{red}{F} \textcolor{red}{l_3},$	$+ i * i\#, 6)$	Action(3, $+$) = $r4$, Goto(0, T) = 2
$\vdash (l_0 \textcolor{red}{T} \textcolor{red}{l_2},$	$+ i * i\#, 64)$	Action(2, $+$) = $r2$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$+ i * i\#, 642)$	Action(1, $+$) = $s6$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6},$	$i * i\#, 642)$	Action(6, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{i} \textcolor{red}{l_5},$	$* i\#, 642)$	Action(5, i) = $r6$, Goto(6, F) = 3
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{F} \textcolor{red}{l_3},$	$* i\#, 6426)$	Action(3, $*$) = $r4$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$* i\#, 64264)$	Action(9, $*$) = $s7$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$i\#, 64264)$	Action(7, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$\#, 64264)$	Action(5, $\#$) = $r6$, Goto(7, F) = 10
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{F} \textcolor{red}{l_{10}},$	$\#, 642646)$	Action(10, $\#$) = $r3$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$\#, 6426463)$	Action(9, $\#$) = $r1$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$\#, 64264631)$	Action(1, $\#$) = acc
$\vdash (acc,$	$\#, 64264631)$	

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G_{Arith} の PDT による解析

$(l_0,$	$i + i * i\#, \varepsilon)$	Action(0, i) = $s5$
$\vdash (l_0 \textcolor{red}{i} \textcolor{red}{l_5},$	$+ i * i\#, \varepsilon)$	Action(5, $+$) = $r6$, Goto(0, F) = 3
$\vdash (l_0 \textcolor{red}{F} \textcolor{red}{l_3},$	$+ i * i\#, 6)$	Action(3, $+$) = $r4$, Goto(0, T) = 2
$\vdash (l_0 \textcolor{red}{T} \textcolor{red}{l_2},$	$+ i * i\#, 64)$	Action(2, $+$) = $r2$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$+ i * i\#, 642)$	Action(1, $+$) = $s6$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6},$	$i * i\#, 642)$	Action(6, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{i} \textcolor{red}{l_5},$	$* i\#, 642)$	Action(5, i) = $r6$, Goto(6, F) = 3
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{F} \textcolor{red}{l_3},$	$* i\#, 6426)$	Action(3, $*$) = $r4$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$* i\#, 64264)$	Action(9, $*$) = $s7$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7},$	$i\#, 64264)$	Action(7, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$\#, 64264)$	Action(5, $\#$) = $r6$, Goto(7, F) = 10
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{F} \textcolor{red}{l_{10}},$	$\#, 642646)$	Action(10, $\#$) = $r3$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$\#, 6426463)$	Action(9, $\#$) = $r1$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$\#, 64264631)$	Action(1, $\#$) = acc
$\vdash (acc,$	$\cdot, 64264631)$	

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G_{Arith} の PDT による解析

$(l_0,$	$i + i * i\#, \varepsilon)$	Action(0, i) = $s5$
$\vdash (l_0$ i $l_5,$	$+ i * i\#, \varepsilon)$	Action(5, $+$) = $r6$, Goto(0, F) = 3
$\vdash (l_0$ F $l_3,$	$+ i * i\#, 6)$	Action(3, $+$) = $r4$, Goto(0, T) = 2
$\vdash (l_0$ T $l_2,$	$+ i * i\#, 64)$	Action(2, $+$) = $r2$, Goto(0, E) = 1
$\vdash (l_0$ E $l_1,$	$+ i * i\#, 642)$	Action(1, $+$) = $s6$
$\vdash (l_0$ E $l_1 + l_6,$	$i * i\#, 642)$	Action(6, i) = $s5$
$\vdash (l_0$ E $l_1 + l_6$ i $l_5,$	$* i\#, 642)$	Action(5, i) = $r6$, Goto(6, F) = 3
$\vdash (l_0$ E $l_1 + l_6$ F $l_3,$	$* i\#, 6426)$	Action(3, $*$) = $r4$, Goto(6, T) = 9
$\vdash (l_0$ E $l_1 + l_6$ T $l_9,$	$* i\#, 64264)$	Action(9, $*$) = $s7$
$\vdash (l_0$ E $l_1 + l_6$ T $l_9 * l_7,$	$i\#, 64264)$	Action(7, i) = $s5$
$\vdash (l_0$ E $l_1 + l_6$ T $l_9 * l_7$ i $l_5,$	$\#, 64264)$	Action(5, $\#$) = $r6$, Goto(7, F) = 10
$\vdash (l_0$ E $l_1 + l_6$ T $l_9 * l_7$ F $l_{10},$	$\#, 642646)$	Action(10, $\#$) = $r3$, Goto(6, T) = 9
$\vdash (l_0$ E $l_1 + l_6$ T $l_9,$	$\#, 6426463)$	Action(9, $\#$) = $r1$, Goto(0, E) = 1
$\vdash (l_0$ E $l_1,$	$\#, 64264631)$	Action(1, $\#$) = acc
$\vdash (acc,$	$\#, 64264631)$	

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G_{Arith} の PDT による解析

$(l_0,$	$i + i * i\#, \varepsilon)$	Action(0, i) = $s5$
$\vdash (l_0 \textcolor{red}{i} \textcolor{red}{l_5},$	$+ i * i\#, \varepsilon)$	Action(5, $+$) = $r6$, Goto(0, F) = 3
$\vdash (l_0 \textcolor{red}{F} \textcolor{red}{l_3},$	$+ i * i\#, 6)$	Action(3, $+$) = $r4$, Goto(0, T) = 2
$\vdash (l_0 \textcolor{red}{T} \textcolor{red}{l_2},$	$+ i * i\#, 64)$	Action(2, $+$) = $r2$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$+ i * i\#, 642)$	Action(1, $+$) = $s6$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6},$	$i * i\#, 642)$	Action(6, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{i} \textcolor{red}{l_5},$	$* i\#, 642)$	Action(5, i) = $r6$, Goto(6, F) = 3
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{F} \textcolor{red}{l_3},$	$* i\#, 6426)$	Action(3, $*$) = $r4$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$* i\#, 64264)$	Action(9, $*$) = $s7$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7},$	$i\#, 64264)$	Action(7, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$\#, 64264)$	Action(5, $\#$) = $r6$, Goto(7, F) = 10
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{F} \textcolor{red}{l_{10}},$	$\#, 642646)$	Action(10, $\#$) = $r3$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$\#, 6426463)$	Action(9, $\#$) = $r1$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$\#, 64264631)$	Action(1, $\#$) = acc
$\vdash (acc,$	$\#, 64264631)$	

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G_{Arith} の PDT による解析

$(l_0,$	$i + i * i\#, \varepsilon)$	Action(0, i) = $s5$
$\vdash (l_0 \textcolor{red}{i} \textcolor{red}{l_5},$	$+ i * i\#, \varepsilon)$	Action(5, $+$) = $r6$, Goto(0, F) = 3
$\vdash (l_0 \textcolor{red}{F} \textcolor{red}{l_3},$	$+ i * i\#, 6)$	Action(3, $+$) = $r4$, Goto(0, T) = 2
$\vdash (l_0 \textcolor{red}{T} \textcolor{red}{l_2},$	$+ i * i\#, 64)$	Action(2, $+$) = $r2$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$+ i * i\#, 642)$	Action(1, $+$) = $s6$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6},$	$i * i\#, 642)$	Action(6, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{i} \textcolor{red}{l_5},$	$* i\#, 642)$	Action(5, i) = $r6$, Goto(6, F) = 3
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{F} \textcolor{red}{l_3},$	$* i\#, 6426)$	Action(3, $*$) = $r4$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$* i\#, 64264)$	Action(9, $*$) = $s7$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7},$	$i\#, 64264)$	Action(7, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$\#, 64264)$	Action(5, $\#$) = $r6$, Goto(7, F) = 10
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{F} \textcolor{red}{l_{10}},$	$\#, 642646)$	Action(10, $\#$) = $r3$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$\#, 6426463)$	Action(9, $\#$) = $r1$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$\#, 64264631)$	Action(1, $\#$) = acc
$\vdash (acc,$	$\#, 64264631)$	

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G_{Arith} の PDT による解析

$(l_0,$	$i + i * i\#, \varepsilon)$	Action(0, i) = $s5$
$\vdash (l_0 \text{ i } l_5,$	$+ i * i\#, \varepsilon)$	Action(5, $+$) = $r6$, Goto(0, F) = 3
$\vdash (l_0 \text{ F } l_3,$	$+ i * i\#, 6)$	Action(3, $+$) = $r4$, Goto(0, T) = 2
$\vdash (l_0 \text{ T } l_2,$	$+ i * i\#, 64)$	Action(2, $+$) = $r2$, Goto(0, E) = 1
$\vdash (l_0 \text{ E } l_1,$	$+ i * i\#, 642)$	Action(1, $+$) = $s6$
$\vdash (l_0 \text{ E } l_1 + l_6,$	$i * i\#, 642)$	Action(6, i) = $s5$
$\vdash (l_0 \text{ E } l_1 + l_6 \text{ i } l_5,$	$* i\#, 642)$	Action(5, i) = $r6$, Goto(6, F) = 3
$\vdash (l_0 \text{ E } l_1 + l_6 \text{ F } l_3,$	$* i\#, 6426)$	Action(3, $*$) = $r4$, Goto(6, T) = 9
$\vdash (l_0 \text{ E } l_1 + l_6 \text{ T } l_9,$	$* i\#, 64264)$	Action(9, $*$) = $s7$
$\vdash (l_0 \text{ E } l_1 + l_6 \text{ T } l_9 * l_7,$	$i\#, 64264)$	Action(7, i) = $s5$
$\vdash (l_0 \text{ E } l_1 + l_6 \text{ T } l_9 * l_7 \text{ i } l_5,$	$\#, 64264)$	Action(5, $\#$) = $r6$, Goto(7, F) = 10
$\vdash (l_0 \text{ E } l_1 + l_6 \text{ T } l_9 * l_7 \text{ F } l_{10},$	$\#, 642646)$	Action(10, $\#$) = $r3$, Goto(6, T) = 9
$\vdash (l_0 \text{ E } l_1 + l_6 \text{ T } l_9,$	$\#, 6426463)$	Action(9, $\#$) = $r1$, Goto(0, E) = 1
$\vdash (l_0 \text{ E } l_1,$	$\#, 64264631)$	Action(1, $\#$) = acc
$\vdash (acc,$	$\#, 64264631)$	

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G_{Arith} の PDT による解析

$(l_0,$	$i + i * i\#, \varepsilon)$	Action(0, i) = $s5$
$\vdash (l_0 \textcolor{red}{i} \textcolor{red}{l_5},$	$+ i * i\#, \varepsilon)$	Action(5, $+$) = $r6$, Goto(0, F) = 3
$\vdash (l_0 \textcolor{red}{F} \textcolor{red}{l_3},$	$+ i * i\#, 6)$	Action(3, $+$) = $r4$, Goto(0, T) = 2
$\vdash (l_0 \textcolor{red}{T} \textcolor{red}{l_2},$	$+ i * i\#, 64)$	Action(2, $+$) = $r2$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$+ i * i\#, 642)$	Action(1, $+$) = $s6$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6},$	$i * i\#, 642)$	Action(6, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{i} \textcolor{red}{l_5},$	$* i\#, 642)$	Action(5, i) = $r6$, Goto(6, F) = 3
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{F} \textcolor{red}{l_3},$	$* i\#, 6426)$	Action(3, $*$) = $r4$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$* i\#, 64264)$	Action(9, $*$) = $s7$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$i\#, 64264)$	Action(7, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{F} \textcolor{red}{l_{10}},$	$\#, 64264)$	Action(5, $\#$) = $r6$, Goto(7, F) = 10
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{F} \textcolor{red}{l_{10}},$	$\#, 6426463)$	Action(10, $\#$) = $r3$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$\#, 64264631)$	Action(9, $\#$) = $r1$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$\#, 64264631)$	Action(1, $\#$) = acc
$\vdash (acc,$	$\#, 64264631)$	

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$(l_0,$	$i + i * i\#, \varepsilon)$	Action(0, i) = $s5$
$\vdash (l_0 \textcolor{red}{i} \textcolor{red}{l_5},$	$+i * i\#, \varepsilon)$	Action(5, $+$) = $r6$, Goto(0, F) = 3
$\vdash (l_0 \textcolor{red}{F} \textcolor{red}{l_3},$	$+i * i\#, 6)$	Action(3, $+$) = $r4$, Goto(0, T) = 2
$\vdash (l_0 \textcolor{red}{T} \textcolor{red}{l_2},$	$+i * i\#, 64)$	Action(2, $+$) = $r2$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$+i * i\#, 642)$	Action(1, $+$) = $s6$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6},$	$i * i\#, 642)$	Action(6, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{i} \textcolor{red}{l_5},$	$*i\#, 642)$	Action(5, i) = $r6$, Goto(6, F) = 3
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{F} \textcolor{red}{l_3},$	$*i\#, 6426)$	Action(3, $*$) = $r4$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$*i\#, 64264)$	Action(9, $*$) = $s7$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$i\#, 64264)$	Action(7, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$\#, 64264)$	Action(5, $\#$) = $r6$, Goto(7, F) = 10
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{F} \textcolor{red}{l_{10}},$	$\#, 642646)$	Action(10, $\#$) = $r3$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$\#, 6426463)$	Action(9, $\#$) = $r1$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$\#, 64264631)$	Action(1, $\#$) = acc
$\vdash (acc,$	$\#, 64264631)$	

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G_{Arith} の PDT による解析

$(l_0,$	$i + i * i\#, \varepsilon)$	Action(0, i) = $s5$
$\vdash (l_0 \textcolor{red}{i} \textcolor{red}{l_5},$	$+i * i\#, \varepsilon)$	Action(5, $+$) = $r6$, Goto(0, F) = 3
$\vdash (l_0 \textcolor{red}{F} \textcolor{red}{l_3},$	$+i * i\#, 6)$	Action(3, $+$) = $r4$, Goto(0, T) = 2
$\vdash (l_0 \textcolor{red}{T} \textcolor{red}{l_2},$	$+i * i\#, 64)$	Action(2, $+$) = $r2$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$+i * i\#, 642)$	Action(1, $+$) = $s6$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6},$	$i * i\#, 642)$	Action(6, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{i} \textcolor{red}{l_5},$	$*i\#, 642)$	Action(5, i) = $r6$, Goto(6, F) = 3
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{F} \textcolor{red}{l_3},$	$*i\#, 6426)$	Action(3, $*$) = $r4$, Goto(6, T) = 9
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$*i\#, 64264)$	Action(9, $*$) = $s7$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$i\#, 64264)$	Action(7, i) = $s5$
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9} * \textcolor{red}{l_7} \textcolor{red}{i} \textcolor{red}{l_5},$	$\#, 64264)$	Action(5, $\#$) = $r6$, Goto(7, F) = 10
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$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1} + \textcolor{red}{l_6} \textcolor{red}{T} \textcolor{red}{l_9},$	$\#, 6426463)$	Action(9, $\#$) = $r1$, Goto(0, E) = 1
$\vdash (l_0 \textcolor{red}{E} \textcolor{red}{l_1},$	$\#, 64264631)$	Action(1, $\#$) = acc
$\vdash (acc,$	$\#, 64264631)$	

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G_{Arith} の PDT による解析

$(l_0,$	$i + i * i\#, \varepsilon)$	Action(0, i) = $s5$
$\vdash (l_0$ i $l_5,$	$+ i * i\#, \varepsilon)$	Action(5, $+$) = $r6$, Goto(0, F) = 3
$\vdash (l_0$ F $l_3,$	$+ i * i\#, 6)$	Action(3, $+$) = $r4$, Goto(0, T) = 2
$\vdash (l_0$ T $l_2,$	$+ i * i\#, 64)$	Action(2, $+$) = $r2$, Goto(0, E) = 1
$\vdash (l_0$ E $l_1,$	$+ i * i\#, 642)$	Action(1, $+$) = $s6$
$\vdash (l_0$ E $l_1 + l_6,$	$i * i\#, 642)$	Action(6, i) = $s5$
$\vdash (l_0$ E $l_1 + l_6$ i $l_5,$	$* i\#, 642)$	Action(5, i) = $r6$, Goto(6, F) = 3
$\vdash (l_0$ E $l_1 + l_6$ F $l_3,$	$* i\#, 6426)$	Action(3, $*$) = $r4$, Goto(6, T) = 9
$\vdash (l_0$ E $l_1 + l_6$ T $l_9,$	$* i\#, 64264)$	Action(9, $*$) = $s7$
$\vdash (l_0$ E $l_1 + l_6$ T $l_9 * l_7$ i $l_5,$	$i\#, 64264)$	Action(7, i) = $s5$
$\vdash (l_0$ E $l_1 + l_6$ T $l_9 * l_7$ i $l_5,$	$\#, 64264)$	Action(5, $\#$) = $r6$, Goto(7, F) = 10
$\vdash (l_0$ E $l_1 + l_6$ T $l_9 * l_7$ F $l_{10},$	$\#, 642646)$	Action(10, $\#$) = $r3$, Goto(6, T) = 9
$\vdash (l_0$ E $l_1 + l_6$ T $l_9,$	$\#, 6426463)$	Action(9, $\#$) = $r1$, Goto(0, E) = 1
$\vdash (l_0$ E $l_1,$	$\#, 64264631)$	Action(1, $\#$) = acc
$\vdash (acc,$	$', 64264631)$	

0: $E' \rightarrow E$

1: $E \rightarrow E + T$ 2: $E \rightarrow T$

3: $T \rightarrow T * F$ 4: $T \rightarrow F$

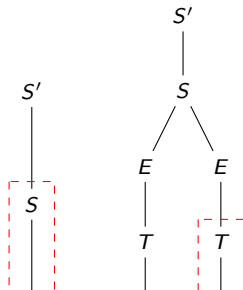
5: $F \rightarrow (E)$ 6: $F \rightarrow i$

SLR(1) 構文解析ができない文法

$$G_{=} = \langle \{S, E, T\}, \{=, +, i\}, P_{=}, S \rangle$$

の開始記号を S' とした拡大文法 $G'_{=}$

$$P'_{=} = \left\{ \begin{array}{lll} (0) & S' \rightarrow S\# & (1) \quad S \rightarrow E = E \quad (2) \quad S \rightarrow i \\ (3) & E \rightarrow E + T & (4) \quad E \rightarrow T \quad (5) \quad T \rightarrow i \end{array} \right\}$$



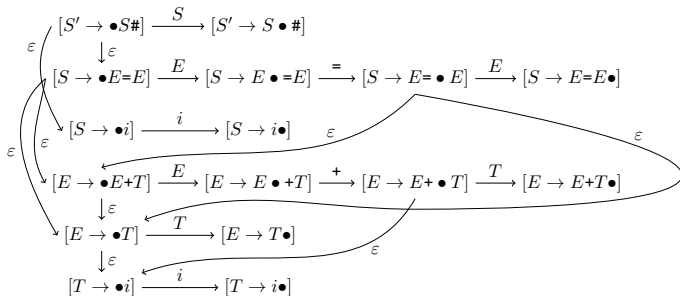
$i = i$ を解析するとき、 $T \rightarrow i$ と $S \rightarrow i$ を
区別できない。

$S \rightarrow i$ の還元の可能性を Follow_1 のみで排
除できない。

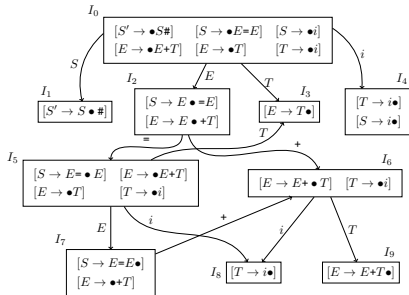
(実際はありえない)

SLR(1) ではない

$G_{=}$ の正準 LR(0) 遷移系



$G_{=}$ の正準 LR(0) 遷移系



$G_{=}$ の SLR 構文解析表

	action				Goto		
	i	$=$	$+$	$\#$	S	E	T
0	s4				1	2	3
1				acc			
2	s5	s6					
3	r4	r4	r4				
4	r5	r5	r5	r2			
5	s8					7	3
6	s8						9
7		s6	r1				
8	r5	r5	r5				
9	r3	r3	r3				

状態 4 の reduce/reduce conflict

$\#$ が先読み文字のときに S に還元するか T に還元するかが不明